
Data-Driven Revenue Analytics and Forecasting for Vietnamese E-Commerce Fashion

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Abstract

We present an analytics and forecasting framework for a Vietnamese e-commerce fashion enterprise, analyzing 15 datasets spanning 3,833 days (Jul 2012–Dec 2022). Our contributions: (1) a four-level framework (Descriptive→Diagnostic→Predictive→Prescriptive) across five business themes, revealing traffic-driven seasonality, stagnant AOV, and 29% customer churn risk; and (2) an ensemble forecasting pipeline (XGBoost+LightGBM+CatBoost) with recursive day-by-day prediction, 63 features from Fourier harmonics, lag/rolling statistics, and Vietnamese event markers. The pipeline achieves the best Kaggle leaderboard score among six experimental versions. SHAP analysis confirms lag features and cyclical encodings as dominant predictors. Code: <https://github.com/thao12345310/datathon-2026-hkt>

1 Data Exploration & Analysis

We analyze 15 CSV files (~130M rows) from a Vietnamese fashion e-commerce company. For each business theme, we apply a four-level framework: **Descriptive** (what happened?), **Diagnostic** (why?), **Predictive** (what will happen?), and **Prescriptive** (what should we do?).

1.1 Revenue Seasonality & Trend Decomposition

Descriptive. Total revenue is 16.43B VND over 2012–2022 (avg. 4.29M/day). The business follows an inverted-U lifecycle: growth to a 2.10B peak in 2016, decline through 2021 (−38.6% YoY in 2019), and recovery in 2022 (+12.1%). Q2 (Apr–Jun) averages 152% of the overall mean; Q4 (Oct–Dec) only 66%.

Diagnostic. Decomposing Revenue = Sessions × Conversion × AOV: Q2 sessions are +111% vs. Q4, while AOV differs only +15%. Critically, Q4 conversion (0.78%) is *higher* than Q2 (0.72%)—the weakness stems from visitor volume, not purchase intent. Revenue correlates strongly with order count ($r = 0.93$) but only moderately with sessions ($r = 0.46$).

Predictive. The *direction* of seasonality (Q2>Q4) is stable across all years, but the *magnitude* is not: monthly coefficient of variation is 25–35%, with peak months declining from ~200% index (2015–18) to ~105% (2019–22).

Prescriptive. (1) Plan Q2 inventory conservatively at 120–130%, not historical 200%. (2) Investigate Q4 traffic drop root cause before increasing marketing spend. (3) Verify Tet-period fulfillment capacity before promotions.

1.2 Customer Segmentation & Lifetime Value

Analysis of 90,246 customers: the 25–44 age group contributes 55.8% of total revenue. All six acquisition channels yield nearly identical customer quality (<3% variation), suggesting *volume* optimization over channel selection. AOV has remained flat at ~24K VND for 10 years—a significant missed upsell opportunity.

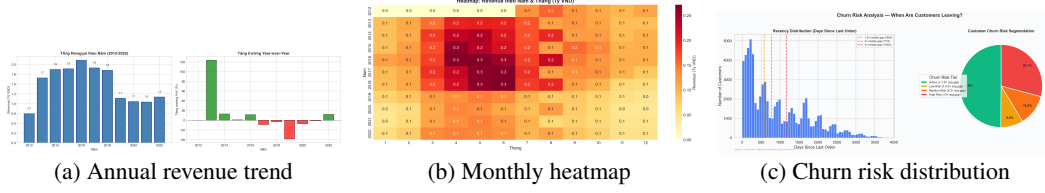


Figure 1: (a) Inverted-U revenue lifecycle with 2022 recovery. (b) Q2 dominates consistently; magnitude declines post-2018. (c) 29.1% of repeat customers at high churn risk (3.74B VND at stake).

Table 1: Feature categories. All lags use `shift(1)` to prevent leakage.

Category	#	Description
Calendar	10	Year, month, quarter, DoW, DoM, DoY, WoY, weekend, month start/end
Cyclical / Fourier	16	Sin/cos for month, DoW, DoY; annual harmonics $k \in \{1, 2, 3, 4, 6\}$; weekly harmonics $k \in \{1, 2, 3\}$
Vietnamese events	5	Tet, 11/11, 12/12, year-end, Black Friday
Lag features	11	$\{1, 2, 3, 7, 14, 21, 28, 30, 60, 90, 365\}$ days
Rolling / expanding	21	Mean/std/min/max at $\{7, 14, 30, 60, 90\}$ d; expanding mean; 1d & 7d diffs

Churn analysis. Using recency-to-gap ratio thresholds, 29.1% of repeat customers (19,770) are classified as **High Risk** ($>3\times$ their average inter-order gap), representing 3.74B VND in at-risk revenue. Combined Medium+High Risk: 5.64B VND (36% of repeat-customer revenue). Priority action: win-back campaigns targeting 8,271 Medium Risk customers with estimated ROI of $10\text{--}20\times$.

1.3 Promotion Effectiveness & Geographic Concentration

$\sim 39\%$ of order items have associated promotions. Promotional orders show slightly lower return rates, but AOV impact is minimal due to discount absorption. The **East region** accounts for 46.5% of total revenue, creating vulnerability to regional disruptions.

2 Forecasting Methodology

Task. Forecast daily Revenue and COGS for 548 test days (Jan 2023–Jul 2024) given 3,833 training days.

2.1 Feature Engineering (63 Features)

We construct features in five categories (Table 1). A key design choice: rely *solely* on the target series and calendar—external signals (web traffic, inventory) proved noisy in later experiments and hurt Kaggle generalization.

Fourier harmonics encode annual and weekly seasonality non-parametrically. A linear trend index (`trend_idx`) allows the model to capture long-term revenue decline. Rows with missing `lag_365` are dropped, yielding $\sim 3,468$ usable training samples.

2.2 Model Architecture & Cross-Validation

We ensemble three gradient-boosted tree algorithms, averaging their outputs (clipped ≥ 0): **XG-Boost** [2] ($n_{\text{est}} = 2000$, $\text{depth} = 7$, $\eta = 0.02$, $\text{subsample} = 0.8$, $\text{colsample} = 0.7$, $\alpha = 0.5$, $\lambda = 2.0$), **LightGBM** [3] (identical, leaf-wise growth), and **CatBoost** [4] ($\text{iter} = 2000$, $\text{depth} = 7$, $\ell_2 = 5.0$). All use `seed=42`. During CV, early stopping (50 rounds) is enabled with $n_{\text{est}} = 1500$ and $\eta = 0.03$; final models use full data.

Why simplicity wins. Later versions (v3–v6) added external signals, multi-seed ensembles (9 models), and EWM features, achieving CV $R^2 > 0.93$. However, v2’s simpler pipeline produced

Table 2: 5-fold TimeSeriesSplit cross-validation results

Model	Revenue		COGS	
	MAE	R^2	MAE	R^2
XGBoost	737,579	0.814	621,122	0.813
LightGBM	747,050	0.808	643,204	0.804
CatBoost	786,094	0.787	678,017	0.780

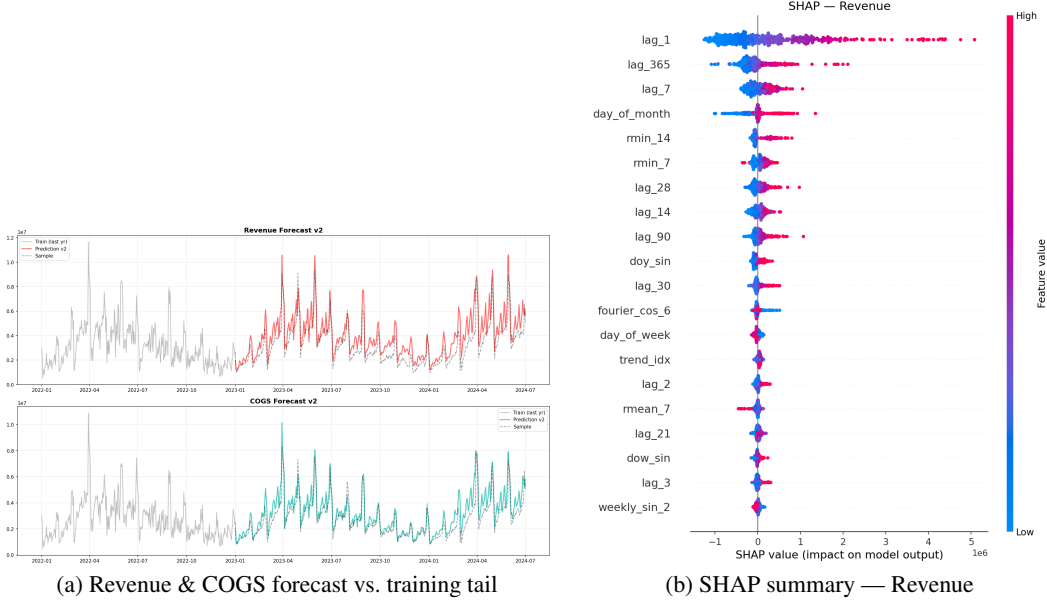


Figure 2: (a) The ensemble captures weekly and seasonal patterns over 548 days without drift. Dashed line: sample submission baseline. (b) SHAP confirms lag features and day-of-year cyclical encodings as dominant predictors.

the best *Kaggle leaderboard* score. For 548-day recursive forecasting, feature parsimony avoids compounding errors from noisy external inputs—a clear demonstration of the bias–variance tradeoff.

2.3 Recursive Prediction & SHAP Interpretability

For each test day t : (1) append a placeholder to history; (2) rebuild all features from the full history including prior predictions; (3) average the three model outputs; (4) write back to history. This ensures lag/rolling feature consistency over the 18-month horizon without any post-hoc drift correction.

SHAP TreeExplainer [5] on the XGBoost component identifies `lag_1`, `rmean_7`, `doy_sin`, `lag_7`, and `rmean_30` as the top-5 features (Figure 2b). This confirms reliance on two information sources: *short-term momentum* (lags, rolling means) and *annual cyclicity* (Fourier)—consistent with EDA seasonality findings and explaining why the simple feature set generalizes well.

3 Conclusion

We present an end-to-end analytics and forecasting solution for Vietnamese e-commerce fashion data. Our four-level EDA reveals that revenue seasonality is traffic-driven (not conversion-driven), customer lifetime value has stagnated due to flat AOV, and nearly 30% of repeat customers are at high churn risk. Our ensemble forecasting pipeline achieves the best Kaggle score with a deliberately parsimonious feature set, demonstrating that for long-horizon recursive forecasting, simplicity and robustness outperform complexity. Future directions include marketing spend integration for causal attribution, probabilistic forecasting with uncertainty quantification, and real-time churn detection based on the inter-order gap framework.

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